Ant Colony Module Homework 7

Due April 17 Midnight

For the 15 department QAP from Nugent you already studied use an ACO and do the following steps.

Ant Colony Optimization

1. Code a simple ant colony method using global pheromone updating only. Carefully devise an encoding and a local heuristic.

2. Run your Ant Colony.

3. Perform the following changes on your ant colony code (one by one) and compare the results.

   - change the initial starting solutions (initial population members) 4 more times (for a total of 5 seeds)
   - change the population size - smaller and larger than your original choice and run for 5 times each
   - change the initial pheromone amount for the original population size – smaller and larger than your original choice and run for the same 5 initial starting populations as in bullet 1
   - change the value of ρ and run for the best combination of population size and pheromone from above for the same 5 initial starting populations as in the first bullet
   - change α to 0 (this creates a search using the local heuristic only) and run for the best combination of population size, pheromone, and ρ from above for the same 5 initial starting populations as in bullet 1.
   - Then change β to 0 (returning α to its original value, this creates a search using the pheromone only) and run for the best combination of population size, pheromone, and ρ from above for the same 5 initial starting populations as in bullet 1
   - add the q_o idea to balance exploitation and exploration search and run for the best combination of population size and pheromone and ρ from above but with the original α and β for the same 5 initial starting populations as in the first bullet
   - add some sort of local pheromone updating (in addition to global updating) and run for the setting of the bullet immediately above for the same 5 initial starting populations as in the first bullet
   - add a more elitist approach to pheromone depositing than your original version of the first bullet and test for the settings from the bullet just above and the same 5 seeds

4. Please include your code with your homework.

This should total 5 + 10 + 10 + 5 + 10 + 5 + 5 + 5 = 55 runs.

This is the sixth of the QAP (quadratic assignment problem) test problems of Nugent et al.* Fifteen departments are to be placed in 15 locations with five departments (columns) in three rows of five departments in each row. The objective is to minimize flow costs between the placed departments. The flow cost is (flow * distance), where both flow and distance are symmetric between any given pair of departments. Below is the flow (lower half) between departments and rectilinear distance (upper half) between locations matrix. The optimal solution is 575 (or 1150 if you double the flows).

* C. E. Nugent, T. E. Vollmann and J. Ruml, An experimental comparison of techniques for the assignment of facilities to locations. *Operations Research* 16, 150-173 (1968)

Dept	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	-	1	2	3	4	1	2	3	4	5	2	3	4	5	6
2	10	-	1	2	3	2	1	2	3	4	3	2	3	4	5
3	0	1	-	1	2	3	2	1	2	3	4	3	2	3	4
4	5	3	10	-	1	4	3	2	1	2	5	4	3	2	3
5	1	2	2	1	-	5	4	3	2	1	6	5	4	3	2
6	0	2	0	1	3	-	1	2	3	4	1	2	3	4	5
7	1	2	2	5	5	2	-	1	2	3	2	1	2	3	4
8	2	3	5	0	5	2	6	-	1	2	3	2	1	2	3
9	2	2	4	0	5	1	0	5	-	1	4	3	2	1	2
10	2	0	5	2	1	5	1	2	0	-	5	4	3	2	1
11	2	2	2	1	0	0	5	10	10	0	-	1	2	3	4
12	0	0	2	0	3	0	5	0	5	4	5	-	1	2	3
13	4	10	5	2	0	2	5	5	10	0	0	3	-	1	2
14	0	5	5	5	5	5	1	0	0	0	5	3	10	-	1
15	0	0	5	0	5	10	0	0	2	5	0	0	2	4	-

Extra Credit – up to 25 more points

Devise an encoding and local heuristic for the Orienteering Problem. Using your encoding and local heuristic code a simple ACO for the orienteering problem. Solve this 33 city problem below. Run for at least 5 seeds for each of five different variations of the ACO (your choice of variations) and report the results.

****
* OP test instances *
****

The first line contains the following data:

Tmax P

Where
Tmax = available time budget per path
P = number of paths (=1)

The remaining lines contain the data of each point.
For each point, the line contains the following data:

x y S

Where
x = x coordinate
y = y coordinate
S = score

* REMARKS *
- The first point is the starting point.
- The second point is the ending point.
- The Euclidian distance is used.
- The last number is the is the prize or value (S)

This is a 33-city problem with a best-known solution of 550.

55	1	
19.1	24.3	0 Starting point
18.2	24.0	0 Ending point
12.6	24.9	20
14.4	28.0	20
16.9	28.1	20
20.7	28.2	20
12.5	26.6	20
21.8	27.3	20

12.5	22.6	20
22.5	17.0	30
19.9	15.0	30
14.9	15.1	30
11.5	18.6	30
12.4	29.8	30
17.8	28.1	30
9.1	29.8	40
10.0	32.6	40
13.9	33.1	40
19.95	10.3	40
15.2	8.0	40
14.7	31.2	50
7.4	36.5	50
21.0	25.5	50
18.0	25.3	10
19.5	24.7	10
21.4	21.8	10
16.0	21.4	10
18.65	26.2	10
17.9	28.9	10
14.3	19.9	20
17.0	19.0	20
10.80	21.0	20
15.7	23.7	10